

Predicted Antibiotic Resistance Phenotype of E. coli QA5221 (CARD Database, n=54 genes)								Predicted Phenotype
Antibiotic Class	Acquired Resistance Genes			Intrinsic Efflux/Regulatory				
Beta-lactam	TEM-1			acrA	acrB	ToIC		RESISTANT
Aminoglycoside	aph3'la	aadA	aadA2	acrD				RESISTANT
Sulfonamide	sul1	sul3						RESISTANT
Trimethoprim	dfrA7							RESISTANT
Phenicol	cmIA1			acrA	acrB			RESISTANT
Macrolide	estX			acrA	acrB	mdfA		RESISTANT
Tetracycline	tet(A)			acrA	acrB	mdfA		RESISTANT
Biocide/QAC	qacL	qacEdelta1		acrA	acrB			RESISTANT
Fluoroquinolone				acrA	acrB	acrEF	emrAB	INTERMEDIATE
Carbapenem				acrA	acrB	ToIC		SUSCEPTIBLE
Polymyxin/Peptide				pmrF	eptA			INTERMEDIATE
Aminocoumarin				ToIC	mdtABC			INTERMEDIATE
Rifamycin				acrA	acrB	ToIC		INTERMEDIATE
Glycylcycline				acrA	acrB			INTERMEDIATE

Acquired resistance gene (mobile element)

Intrinsic efflux pump / regulator

Intermediate — phenotype depends on expression